

RUIHONG MAI

Machine Learning | Deep Learning | Generative Modeling

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EDUCATION

Beijing Institute of Technology, Zhuhai

Undergraduate studies in Data Science and Big Data Technology

GPA: 85.333/100 | **Honors:** Four consecutive college scholarships

Selected coursework (scores out of 100): Machine Learning (90), Pattern Recognition (92), Advanced Topics in Deep Learning (91), Probability and Statistics (92), Advanced Calculus (91).

RESEARCH & PROJECT EXPERIENCE

Continuous Image Generation with Diffusion Models

Project Lead | Jul 2024 - May 2025

- Investigated text-driven, multi-frame image generation, focusing on temporal consistency and visual coherence across image sequences.
- Designed an LSTM + Diffusion model to generate visually coherent image sequences.
- Explored BLIP + CLIP and SEED-X multimodal approaches to improve image quality and consistency. The BLIP + CLIP approach reduced FID from **74.63 to 17.40**.

Improving YOLOv5 for Small Object Detection

Project Lead | May 2024 - Jun 2024

- Improved detection of fast-moving, small objects in CS:GO, with a focus on distant player heads; designed smaller anchor boxes and modified the Focus and multi-level BottleneckCSP modules.
- Reported improvements of **1.01% in precision**, **1.93% in recall**, and **1.51% in mAP@0.5**. For head targets, precision and recall improved by **2.89%** and **4.21%**, respectively.

ChatGLM2-6B Fine-Tuning with P-Tuning and LoRA

Project Lead | Nov 2023 - Jan 2024

- Applied P-Tuning and LoRA to fine-tune ChatGLM2-6B on **two NVIDIA RTX 3090 GPUs (24 GB each)**, targeting Chinese dialogue generation and model self-identification tasks.

AWARDS

National Third Prize, Teddy Cup Data Analysis Skills Competition | Team Member

TECHNICAL SKILLS

Programming: Python, R; basic Java

Machine Learning: PyTorch, Transformers, scikit-learn, OpenCV, NumPy

Systems & Data: Linux, Hadoop, Spark, SQL, MySQL; familiarity with Kettle ETL

Methods: Model design, training, deployment and optimization; data mining and visualization